

Lab 11: D3 Scrollytelling

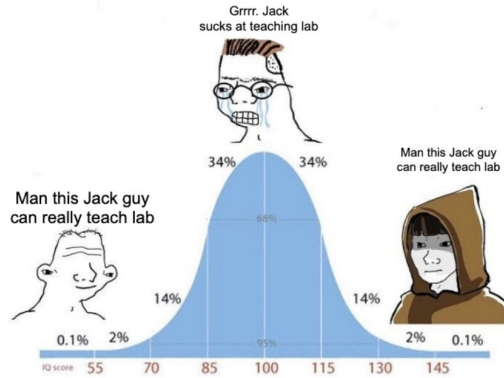
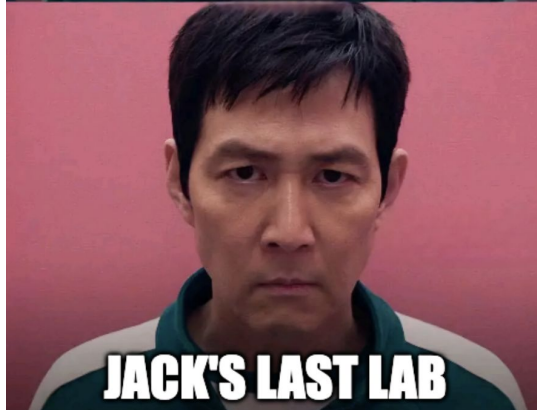
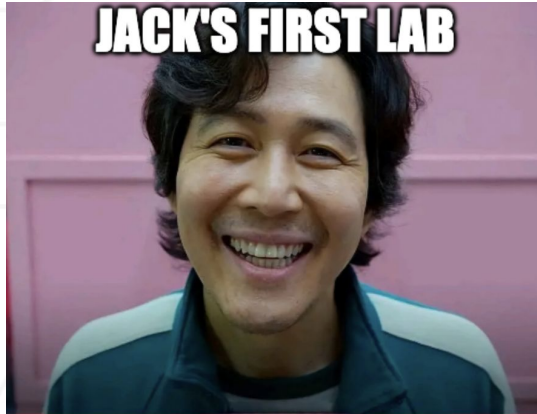
CS 7450: Information Visualization Spring 2026

**Please keep all applause
and tears to the end of
the lab. Thank you!**

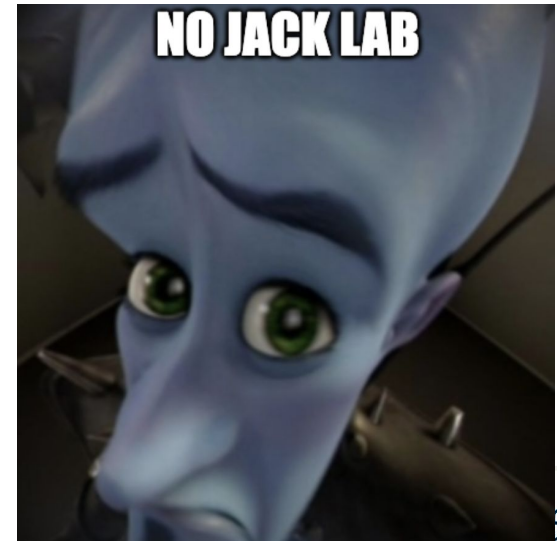
Reminders

- Prototype Part 4 is due this **Friday (04/10)!!!**
- HW4 is due next **Friday (04/17)!!!**
- **CLOS** comes out **April 20th!!!**
- **Monday (4/20)** are going to be TA check in days for Prototypes so please be ready to talk about your projects and answer questions!!!
- Jack and Alysha's office hours will be **canceled** week of **4/20** as you will have in class office hours

Memes



**when I'm in a
TEACHING LAB COMPETITION
competition and my
opponent is JACK**



Goals for the labs

- ~~Learn Web Fundamentals~~
 - ~~HTML, CSS, SVG, JS, DOM, and various tooling methods~~
- ~~D3 Fundamentals~~
 - ~~Data loading, binding, scalings, axes, and various charts~~
- ~~D3 Interactions~~
 - ~~Event listeners, selection, brushing and linking, sorting, and transitions~~
- Advanced D3 Techniques
 - Network visualizations, **scrollytelling**, and geospatial projections

FINAL STRETCH



What is Scrollytelling

- Scrollytelling is a form of interactive visualization where a “story” is told as the user scrolls down the webpage.
 - Sections of story are often accompanied by different visualizations to provide a deeper layer of meaning.
 - It provides more and rich details that allows the users to become more engaged with the story.
- Jack loves Scrollytelling!!!
- Example 1
- Example 2
- Example 3

Scrollama

- Scrollama is a modern, lightweight JavaScript library for scrollytelling
- It uses a **IntersectionObserver** in favor of scroll events
 - Abstracts element-in-view detection and removes the need to monitor scroll events, which contributes to sluggish experiences
- The three main features of Scrollama are:
 - Step triggers: fires an event where an element crosses a visible threshold
 - Step process: fires events more granularly from 0-100% completion of a step
 - Sticky graphic: convenience function for the opinionated (but popular) scrollytelling sticky graphic pattern

Setting up Scrollama

- 1) Include Scrollama in Your Project
- 2) Setup HTML Structure
- 3) Initialize Scrollama
- 4) Configure Scrollama
- 5) Handle Resize
- 6) Implement Your Custom Logic (Story)

Setting up Scrollama

- 1) **Include Scrollama in Your Project**
- 2) Setup HTML Structure
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```
<script src="https://unpkg.com/scrollama"></script>
```

Script to include

Setting up Scrollama

- 1) Include Scrollama in Your Project
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```
<div id="scrolly">
  <div class="sticky">
    <!--Your sticky graphic goes here -->
  </div>
  <div class="scroll">
    <!--Your steps go here -->
    <div class="step" data-step="1">Step 1 content...</div>
    <div class="step" data-step="2">Step 2 content...</div>
    <!--Add more steps as needed -->
  </div>
</div>
```

Basic HTML structure

Setting up Scrollama

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```
// Initialize Scrollama  
const scroller = scrollama();
```

Initialize Scrollama

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```
// Initialize Scrollama
const scroller = scrollama();
scroller.setup({
  step:".step", // class name of the steps
  offset:0.5, // midpoint of the viewport
  debug:true, // optional, shows offset marker for development
})
.onStepEnter(response =>{
  // response = { element, index, direction }
  console.log('Element:', response.element);
  console.log('Index:', response.index);
  console.log('Direction:', response.direction);
  // Do something when a step enters
}).onStepExit(response =>{
  // Do something when a step exits
});
```

Configuration of Scrollama

Setting up Scrollama

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- 5) Handle Resize**
- 6) Implement Your Custom Logic (Story)

```
let stepH = window.innerHeight * 0.8;  
step.style("height", stepH + "px");
```

Handling Resizing

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```
function handleStepEnter(response) {  
  console.log(response);  
  // response = { element, direction, index }  
  // add color to current step only  
  step.classed("is-active", function(d, i) {  
    return i === response.index;  
  });  
  // update graphic based on step  
  switch(response.index) {  
    case 0:  
      firstStep()  
      break;  
    case 1:  
      secondStep()  
      break;  
    case 2:  
      thirdStep()  
      ...  
      break;  
    default:  
      break;  
  }  
}
```

Step Logic

LET'S LIVE CODE!!

